

Tips for ensuring accurate data

Hello, data detective!

Always being curious, enjoying investigation, and committing to tracking down necessary information are all big parts of the data analysis process.

Yet sometimes, that information just isn't as useful as we'd hope!

In this video, you'll gain some great strategies for figuring out how to fix this problem.

Maybe, as a data analyst, you're asked to respond to a business data request.

This is a business question that you can answer with data.

Your **first step** is to **assess the request to make sure you understand what's involved, and the goal or conclusion desired.**

Let's take up this case study.

The request involves a supply chain organization that wants to determine if supply partners have a successful customer experience when purchasing inventory from their online store.

Now, there are a few ways to confirm whether you have the data you need to respond to this request.

You could use analytics software to check the structure of the database.

Then, determine if it is possible to learn when supply partners began searching for a specific product compared to when they clicked the purchase button.

You might also track timestamps on their visits to your website.

This can help confirm that partners are able to efficiently find and buy what they need.

And, it lets you know that they're actively engaging with your website.

Next, you move on to **data cleaning** —sometimes **called data wrangling.**

In this phase, you correct or eliminate any inaccurate or repeat records from the dataset.

For example, maybe a supply partner omits important details from an order, then creates the same order again to correct their mistake.

You'd want to remove the duplicate.

Also, during this phase, you have the opportunity to learn more about your supply partner and define specific segments.

And then, break down the dataset into smaller sections.

Now it's time to check for any outliers or deviations in the dataset.

If a partner mistakenly enters an order for one thousand products, but they really only need ten, you'd want to address that outlier.

Here, you also integrate multiple data sources—or use data from multiple methods or sources—to ensure information is accurate.

In this example, you'd use data from your website, your customer management system, and your purchase orders.

If available, you could also check summary statistics.

You can create these in a dashboard or spreadsheet that calculates the average, mean, median, minimum, maximum, and other various relevant statistics and calculations.

You can also run a quick SQL query that does the same.

Finally, if you've done this type of analysis in the past, it might be helpful to compare the two projects.

If the results vary wildly, this likely indicates that you need to recheck your dataset and your process.

Better yet, consistent results help confirm that you're working with valuable, useful data.

By putting on your detective's cap and viewing data from many different perspectives, you can be sure you're using the best possible data!

Your stakeholders will appreciate your investigative spirit when their requests are fulfilled skillfully and thoroughly!